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VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Department of Mathematics

MODULE 2

INFINITE FOURIER TRANSFORMS

TOPICS COVERED IN THIS MODULE

1. Infinite Fourier Transforms	2. Fourier Cosine & Sine Transforms
3. Inverse Fourier Transforms	4. Properties of Fourier Transforms
5. Discrete Fourier Transform (DFT)	6. Fast Fourier Transform (FFT)

TABLE OF CONTENTS

1	Formulae Reference Sheet	03
2	Complex Fourier Transform (Infinite Fourier Transform)	04
2.1	Definition and Formula	04
2.2	Properties of Fourier Transform	05
2.3	Worked Examples — Fourier Transform	06
3	Fourier Cosine Transform	10
3.1	Definition and Formula	10
3.2	Inverse Fourier Cosine Transform	11
3.3	Worked Examples — Cosine Transform	11
4	Fourier Sine Transform	15
4.1	Definition and Formula	15
4.2	Inverse Fourier Sine Transform	15
4.3	Worked Examples — Sine Transform	16
5	Parseval's Theorem and Integral Equations	20
6	Convolution Theorem	22
7	Self-Reciprocal Functions	24
8	Discrete Fourier Transform (DFT)	25
8.1	Definition and Formula	25
8.2	Matrix Form of DFT	26
8.3	Worked Examples — DFT	27
9	Fast Fourier Transform (FFT)	31
9.1	Decimation in Time (DIT) FFT	32
9.2	Worked Examples — FFT	33
10	Previous Year VTU Questions with Solutions	36
11	Quick Revision Summary	44

1. FORMULAE REFERENCE SHEET

NOTE: Memorize these formulae — they appear directly in every VTU exam!

1.1 Fourier Transform Formulae Summary Table

TYPE	TRANSFORM	INVERSE TRANSFORM
Complex Fourier Transform	$F(u) = \int_{-\infty}^{+\infty} f(x) e^{iux} dx$	$f(x) = (1/2\pi) \int_{-\infty}^{+\infty} F(u) e^{-iux} du$
Fourier Cosine Transform	$F_c(u) = \int_0^{\infty} f(x) \cos(ux) dx$	$f(x) = (2/\pi) \int_0^{\infty} F_c(u) \cos(ux) du$
Fourier Sine Transform	$F_s(u) = \int_0^{\infty} f(x) \sin(ux) dx$	$f(x) = (2/\pi) \int_0^{\infty} F_s(u) \sin(ux) du$

1.2 Key Standard Results

The following results are used repeatedly in problem solving. Commit them to memory.

No.	Formula	Remark
1	$i = \sqrt{-1}, \quad i^2 = -1, \quad i^3 = -i, \quad i^4 = 1$	Powers of i cycle with period 4
2	$1/i = -i$ (multiply num/denom by i)	Rationalization trick
3	$e^{in\theta} = \cos(n\theta) + i \sin(n\theta)$	Euler's formula (positive)
4	$e^{-in\theta} = \cos(n\theta) - i \sin(n\theta)$	Euler's formula (negative)
5	$e^{in\theta} + e^{-in\theta} = 2\cos(n\theta)$	Sum form
6	$e^{in\theta} - e^{-in\theta} = 2i \sin(n\theta)$	Difference form
7	$\int \frac{e^{ax} \cos(bx) dx}{[a^2 + b^2]} = \frac{e^{ax} [a \cos(bx) + b \sin(bx)]}{a^2 + b^2}$	Standard integral
8	$\int \frac{e^{ax} \sin(bx) dx}{[a^2 + b^2]} = \frac{e^{ax} [a \sin(bx) - b \cos(bx)]}{a^2 + b^2}$	Standard integral
9	$\int_0^{\infty} e^{-t^2} dt = \sqrt{\pi}/2$ (Gaussian integral)	Critical for self-reciprocal proof
10	$1 - \cos \theta = 2 \sin^2(\theta/2)$	Trigonometric identity
11	$e^{-\infty} = 0, \quad e^0 = 1$	Boundary conditions at ∞

2. COMPLEX (INFINITE) FOURIER TRANSFORM

The Fourier Transform is one of the most powerful tools in mathematical analysis. It decomposes a function defined on the entire real line into its frequency components. Unlike Fourier series (defined on a finite interval), the Fourier transform works on functions defined over $(-\infty, +\infty)$.

2.1 Definition

Let $f(x)$ be a function defined for all real values of x . If $f(x)$ is absolutely integrable over $(-\infty, +\infty)$ (i.e., $\int |f(x)| dx < \infty$), then the Fourier Transform of $f(x)$ is defined as:

FOURIER
TRANSFORM

$$F(u) = \int_{-\infty}^{+\infty} f(x) \cdot e^{iux} dx$$

The variable u is called the transform variable or frequency variable. The function $F(u)$ is called the Fourier transform (or frequency spectrum) of $f(x)$.

2.1.1 Inverse Fourier Transform

Given the Fourier transform $F(u)$, we can recover the original function $f(x)$ using the Inverse Fourier Transform:

INVERSE FOURIER

$$f(x) = (1/2\pi) \int_{-\infty}^{+\infty} F(u) \cdot e^{-iux} du$$

NOTE: The factor $1/(2\pi)$ comes from the normalization convention used in the definition. Different textbooks may place the $1/(2\pi)$ factor differently, but the product of the two normalization factors is always $1/(2\pi)$.

2.2 Properties of Fourier Transform

Understanding properties helps solve complex problems by breaking them into simpler known results. These are VTU exam favorites.

PROPERTY	STATEMENT	SIGNIFICANCE
Linearity	$F[af(x) + bg(x)] = aF(u) + bG(u)$	Transform of sum = sum of transforms
Scaling	$F[f(ax)] = (1/ a) F(u/a)$	Compression in time = expansion in freq
Time Shifting	$F[f(x-a)] = e^{-iau} F(u)$	Time delay adds phase shift e^{-iau}
Frequency Shifting	$F[e^{iax}f(x)] = F(u+a)$	Multiplication by e^{iax} shifts spectrum
Differentiation (x)	$F[f'(x)] = (-iu) F(u)$	Differentiation $\rightarrow$ multiplication by $(-iu)$

Differentiation (u)	$F[(-ix)^n f(x)] = F^{(n)}(u)$	n-th derivative in freq domain
Integration	$F[\int f(t) dt] = F(u) / (-iu)$	Integration $\rightarrow$ division by $(-iu)$
Conjugate Symmetry	$F[-u] = F^*(u)$ (if $f(x)$ is real)	Real functions have conjugate symmetry
Parseval's Theorem	$\int f(x) ^2 dx = (1/2\pi) \int F(u) ^2 du$	Energy conservation across domains
Convolution	$F[f * g] = F(u) \cdot G(u)$	Convolution in time = product in freq

2.3 Worked Examples — Fourier Transform

Example 1: Find the Fourier Transform of $f(x) = e^{-|x|}$

This is the most fundamental example in Fourier transforms and appears in almost every VTU exam.

SOLUTION

S1

Write the piecewise definition of $f(x) = e^{-|x|}$

Since $|x| = x$ for $x > 0$ and $|x| = -x$ for $x < 0$: $f(x) = e^{-x}$ for $x \geq 0$ and $f(x) = e^x$ for $x < 0$

S2

Apply the Fourier Transform definition

$$F(u) = \int_{-\infty}^{+\infty} f(x) e^{iux} dx = \int_{-\infty}^0 e^x \cdot e^{iux} dx + \int_0^{+\infty} e^{-x} \cdot e^{iux} dx$$

S3

Combine exponents in each integral

$$= \int_{-\infty}^0 e^{x(1+iu)} dx + \int_0^{+\infty} e^{x(-1+iu)} dx = \int_{-\infty}^0 e^{(1+iu)x} dx + \int_0^{+\infty} e^{-(1-iu)x} dx$$

S4

Evaluate each definite integral

First integral: $[e^{(1+iu)x} / (1+iu)]$ from $-\infty$ to $0 = (1/(1+iu))(e^0 - e^{-\infty}) = 1/(1+iu)$
 Second integral: $[e^{-(1-iu)x} / -(1-iu)]$ from 0 to $\infty = (1/(1-iu))(e^0 - e^{-\infty}) = 1/(1-iu)$

S5

Add the two results and simplify

$$F(u) = 1/(1+iu) + 1/(1-iu) = [(1-iu) + (1+iu)] / [(1+iu)(1-iu)] = 2 / (1 - (iu)^2) = 2 / (1 - (-u^2)) = 2 / (1 + u^2)$$

RESULT

$$F(u) = F[e^{-|x|}] = 2 / (1 + u^2)$$

NOTE: Since $e^{-|x|}$ is an EVEN function, its Fourier transform $F(u) = 2/(1+u^2)$ is real and even. This is a general property: even functions have real Fourier transforms.

Example 2: Find Fourier Transform of $f(x) = 1$ for $|x| \leq a$, 0 for $|x| > a$

This is a rectangular pulse (box function). The result gives the sinc function, fundamental in signal processing.

S1**Set up the integral using the piecewise definition**

$$F(u) = \int_{-\infty}^{+\infty} f(x) e^{iux} dx = \int_{-a}^a 0 \cdot e^{iux} dx + \int_{-a}^a 1 \cdot e^{iux} dx + \int_{a}^{+\infty} 0 \cdot e^{iux} dx = \int_{-a}^a e^{iux} dx$$

S2**Integrate and evaluate limits**

$$= [e^{iux}/(iu)] \text{ from } -a \text{ to } +a = (1/iu)[e^{iua} - e^{-iua}] = (1/iu) \cdot 2i \sin(ua) \quad [\text{since } e^{i\theta} - e^{-i\theta} = 2i \sin\theta] = 2\sin(ua)/u$$

RESULT

$$F(u) = 2\sin(au)/u$$

Deducing Integral Values from Fourier Transforms

A very common VTU question type: use the inverse Fourier theorem to evaluate definite integrals. The key idea is:

NOTE: At $x = 0$, the inverse Fourier transform formula gives us the value of the integral involving $F(u)$. This technique converts complex definite integrals into simple substitutions.

From Example 2: $F(u) = 2\sin(au)/u$. Using the inverse Fourier transform with $x = 0$:

$$f(0) = (1/2\pi) \int_{-\infty}^{+\infty} F(u) \cdot e^{i0} du = (1/2\pi) \int_{-\infty}^{+\infty} [2\sin(au)/u] du$$

Since $f(0) = 1$ (from definition, as $0 \leq a$ means $|0| \leq a$):

$$1 = (1/\pi) \int_{-\infty}^{+\infty} \sin(au)/u du = (2/\pi) \int_0^{+\infty} \sin(au)/u du$$

$$\int_0^{+\infty} \sin(au)/u du = \pi/2$$

Putting $a = 1$ and replacing u by x :

CLASSIC RESULT

$$\int_0^{\infty} \sin(x)/x dx = \pi/2$$

IMPORTANT: The result $\int_0^{\infty} \sin(x)/x dx = \pi/2$ is one of the most important definite integrals in mathematics. It appears in signal processing, probability, and physics. VTU has asked this derivation multiple times!

Example 3: Fourier Transform of $f(x) = x \cdot e^{-|x|}$

This is an ODD function. Observe how the final result is purely imaginary — a general property of odd functions.

S1

Piecewise form: $f(x) = xe^x$ ($x < 0$) and $f(x) = xe^{-x}$ ($x \geq 0$)

S2**Split the integral at $x = 0$**

$$F(u) = \int_{-\infty}^0 xe^x \cdot e^{iux} dx + \int_0^{+\infty} xe^{-x} \cdot e^{iux} dx = \int_{-\infty}^0 xe^{(1+iu)x} dx + \int_0^{+\infty} xe^{-(1-iu)x} dx$$

S3	Use integration by parts ($\int u \cdot v' = uv - \int v \cdot u'$ approach) For $\int_{-\infty}^0 x e^{\{(1+iu)x\}} dx$: Let $p = x$, $dq = e^{\{(1+iu)x\}} dx = [x e^{\{(1+iu)x\}} / (1+iu) - e^{\{(1+iu)x\}} / (1+iu)^2]$ from $-\infty$ to $0 = -1/(1+iu)^2$
S4	Similarly for second integral $\int_0^{+\infty} x e^{\{-(1-iu)x\}} dx = [x e^{\{-(1-iu)x\}} / (-(1-iu)) - e^{\{-(1-iu)x\}} / (1-iu)^2]$ from 0 to $\infty = 1/(1-iu)^2$
S5	Add results and simplify $F(u) = -1/(1+iu)^2 + 1/(1-iu)^2 = [-(1-iu)^2 + (1+iu)^2] / [(1+iu)(1-iu)]^2 = [-4iu] / (1+u^2)^2 = 4iu/(1+u^2)^2$
RES ULT	$F(u) = F[x \cdot e^{\{- x \}}] = 4iu/(1+u^2)^2$

NOTE: The transform is purely imaginary ($4iu/\dots$) because $x \cdot e^{\{-|x|\}}$ is an ODD function. This always holds: odd real functions have purely imaginary Fourier transforms.

3. FOURIER COSINE TRANSFORM

The Fourier Cosine Transform is used for EVEN functions defined on $[0, \infty)$. When a function is even ($f(-x) = f(x)$), its Fourier transform simplifies to contain only cosine terms, which gives us the Fourier Cosine Transform. This is also applied when only $x \geq 0$ is specified (the function is defined for positive x only).

3.1 Definition — Fourier Cosine Transform

FOURIER COSINE
TRANSFORM

$$F_c(u) = \int_0^{+\infty} f(x) \cdot \cos(ux) \, dx$$

3.2 Inverse Fourier Cosine Transform

INVERSE COSINE
TRANSFORM

$$f(x) = (2/\pi) \int_0^{+\infty} F_c(u) \cdot \cos(ux) \, du$$

NOTE: The factor $2/\pi$ appears in the inverse because we integrate only from 0 to ∞ (half the real line). Both the function and the kernel $\cos(ux)$ are even, so everything is consistent.

3.3 Worked Examples — Fourier Cosine Transform

Example 1: Find the Fourier Cosine Transform of $f(x) = e^{-ax}$ ($a > 0$)

This is the most standard cosine transform result. The result $F_c(u) = a/(a^2 + u^2)$ and the integral formula derived from it are asked in every VTU exam.

S1

Write the definition

$$F_c(u) = \int_0^{+\infty} f(x) \cdot \cos(ux) \, dx = \int_0^{+\infty} e^{-ax} \cdot \cos(ux) \, dx$$

S2

Apply the standard integral formula

$$\int e^{ax} \cos(bx) \, dx = e^{ax}/(a^2 + b^2) [a \cdot \cos(bx) + b \cdot \sin(bx)] + C$$

Here: replace a with $(-a)$ and b with u : $\int_0^{+\infty} e^{-ax} \cos(ux) \, dx = [e^{-ax}/(a^2 + u^2) (-a \cdot \cos(ux) + u \cdot \sin(ux))] \text{ from } 0 \text{ to } \infty$

S3

Apply boundary conditions

$$\text{At } x = \infty: e^{-\infty} = 0, \text{ so entire expression} = 0$$

$$\text{At } x = 0: e^{-a \cdot 0} = 1, \cos(0) = 1, \sin(0) = 0 \rightarrow 0 - [1/(a^2 + u^2)](-a \cdot 1 + u \cdot 0) = -(-a)/(a^2 + u^2) = a/(a^2 + u^2)$$

RESULT

$$F_c(u) = F_c[e^{-ax}] = a/(a^2 + u^2)$$

Deducing Integral Formula from this Result

Using the Inverse Cosine Transform formula:

$$f(x) = (2/\pi) \int_0^+ \infty Fc(u) \cdot \cos(ux) du = (2/\pi) \int_0^+ \infty [a/(a^2+u^2)] \cdot \cos(ux) du$$

Since $f(x) = e^{-ax}$, we have:

$$e^{-ax} = (2a/\pi) \int_0^+ \infty \cos(ux)/(a^2+u^2) du$$

$$\int_0^+ \infty \cos(ux)/(a^2+u^2) du = (\pi/2a) e^{-ax}$$

Replacing u by x and x by m (renaming dummy variables):

**STANDARD
INTEGRAL**

$$\int_0^+ \infty \cos(mx)/(a^2+x^2) dx = (\pi/2a) e^{-am}$$

IMPORTANT: This integral result $\int_0^+ \infty \cos(mx)/(a^2+x^2) dx = (\pi/2a)e^{-am}$ is **EXTREMELY IMPORTANT** for VTU exams and appears as a direct question or in the deduction part of cosine transform problems.

Example 2: Find the Fourier Cosine Transform of $f(x) = 1 - |x|$ for $|x| \leq 1$, 0 for $|x| > 1$

This is a triangular pulse, a very common VTU problem.

S1	Write the effective range: for $x \geq 0$, $f(x) = 1 - x$ for $0 \leq x \leq 1$, and 0 for $x > 1$
S2	Set up the cosine transform $Fc(u) = \int_0^1 (1-x)\cos(ux) dx + \int_1^+ \infty 0 \cdot \cos(ux) dx$
S3	Integrate by parts: let $p = (1-x)$, $dq = \cos(ux)dx$ $\int_0^1 (1-x)\cos(ux) dx = [(1-x)\sin(ux)/u]_0^1 + (1/u)\int_0^1 \sin(ux) dx = [0 - 0] + (1/u)[- \cos(ux)/u]_0^1 = (1/u^2)[1 - \cos(u)]$
S4	Use the identity $1 - \cos\theta = 2\sin^2(\theta/2)$ $Fc(u) = (1/u^2)(1 - \cos u) = (1/u^2)(2\sin^2(u/2)) = 2\sin^2(u/2)/u^2$
RES ULT	$Fc(u) = 2\sin^2(u/2)/u^2$

Example 3: Fourier Cosine and Sine Transform of $f(x) = 2e^{-3x} + 3e^{-2x}$

This example uses linearity to handle sums of exponentials.

S1	Use Linearity: $Fc[2e^{-3x} + 3e^{-2x}] = 2Fc[e^{-3x}] + 3Fc[e^{-2x}]$
S2	Apply standard result $Fc[e^{-ax}] = a/(a^2+u^2)$ for each term $Fc[e^{-3x}] = 3/(9+u^2)$ and $Fc[e^{-2x}] = 2/(4+u^2)$
S3	Combine using linearity $Fc(u) = 2 \cdot 3/(9+u^2) + 3 \cdot 2/(4+u^2) = 6/(9+u^2) + 6/(4+u^2)$
Cosine Result	$Fc(u) = 6[1/(9+u^2) + 1/(4+u^2)]$

S4**For Fourier SINE Transform: $F_s[e^{-ax}] = u/(a^2+u^2)$**

$$F_s(u) = 2 \cdot u/(9+u^2) + 3 \cdot u/(4+u^2) = 2u/(9+u^2) + 3u/(4+u^2)$$

**Sine
Result**

$$F_s(u) = u[2/(9+u^2) + 3/(4+u^2)]$$

4. FOURIER SINE TRANSFORM

The Fourier Sine Transform is used for ODD functions or for functions defined only on $(0, \infty)$ where we extend them as odd functions to the whole real line. When a function is odd ($f(-x) = -f(x)$), its Fourier transform contains only sine terms.

4.1 Definition — Fourier Sine Transform

FOURIER SINE
TRANSFORM

$$Fs(u) = \int_0^{+\infty} f(x) \cdot \sin(ux) \, dx$$

4.2 Inverse Fourier Sine Transform

INVERSE SINE
TRANSFORM

$$f(x) = (2/\pi) \int_0^{+\infty} Fs(u) \cdot \sin(ux) \, du$$

NOTE: EVEN vs ODD check: If $F(u)$ is the Fourier transform and $f(x)$ is even, then $F_c(u) = F(u)$ and F is real. If $f(x)$ is odd, then $Fs(u)$ relates to the imaginary part of $F(u)$. This is a useful sanity check.

4.3 Worked Examples — Fourier Sine Transform

Example 1: Fourier Sine Transform of $f(x) = e^{-ax}$, $a > 0$

S1	Apply the definition $Fs(u) = \int_0^{+\infty} e^{-ax} \cdot \sin(ux) \, dx$
S2	Use standard integral: $\int e^{ax} \sin(bx) dx = e^{ax}/(a^2+b^2)[a \sin(bx) - b \cos(bx)]$ Here $a \rightarrow -a$, $b \rightarrow u$: $\int_0^{+\infty} e^{-ax} \sin(ux) dx = [e^{-ax}/(a^2+u^2)(-a \sin(ux) - u \cos(ux))] \text{ from } 0 \text{ to } \infty$
S3	Apply limits At $x = \infty$: $e^{-\infty} = 0$ (entire expression $\rightarrow 0$) At $x = 0$: $\sin(0)=0$, $\cos(0)=1 \rightarrow 0 - [1/(a^2+u^2)](0 - u) = u/(a^2+u^2)$
RESULT	$Fs(u) = F_s[e^{-ax}] = u/(a^2 + u^2)$

Example 2: Fourier Sine Transform of $f(x) = e^{-|x|}$ and evaluate $\int_0^{\infty} x \cdot \sin(mx)/(1+x^2) \, dx$

This is a classic VTU problem combining sine transform with integral evaluation.

S1	For $x \geq 0$: $x = x$, so $f(x) = e^{-x}$. Apply the sine transform: $Fs(u) = \int_0^{+\infty} e^{-x} \cdot \sin(ux) \, dx$
S2	This is $Fs[e^{-ax}]$ with $a=1$: $Fs(u) = u/(1+u^2)$
TRANSFORM	$Fs(u) = u/(1 + u^2)$

RES
ULT

Now use the Inverse Fourier Sine Transform to evaluate the integral:

S3	Apply inverse sine transform $f(x) = (2/\pi) \int_0^{\infty} F_s(u) \cdot \sin(ux) du \quad e^{-x} = (2/\pi) \int_0^{\infty} [u/(1+u^2)] \cdot \sin(ux) du$
S4	Multiply both sides by $\pi/2$ $(\pi/2)e^{-x} = \int_0^{\infty} u \cdot \sin(ux)/(1+u^2) du$
S5	Put $x = m$ (rename), then replace dummy variable u by x: $\int_0^{\infty} u \cdot \sin(mu)/(1+u^2) du = (\pi/2)e^{-m}$
INTEGRAL RESULT	$\int_0^{\infty} x \cdot \sin(mx)/(1+x^2) dx = (\pi/2) e^{-m} \quad (m > 0)$

IMPORTANT: This integral result $\int_0^{\infty} x \cdot \sin(mx)/(1+x^2) dx = (\pi/2)e^{-m}$ is directly asked in VTU exams. Always deduce it from the sine transform of e^{-x} .

Example 3: Fourier Sine Transform of $f(x) = e^{-ax}/x$ using Differentiation w.r.t. Parameter

This is an advanced technique — differentiating the transform with respect to u to find the transform of e^{-ax}/x .

S1	We know $F_s[e^{-ax}] = u/(a^2+u^2)$. Now we want $F_s[e^{-ax}/x]$.
S2	Key idea: differentiate F_s w.r.t. u $d/du [F_s(u)] = d/du \left[\int_0^{\infty} (e^{-ax}/x) \cdot \sin(ux) dx \right] = \int_0^{\infty} (e^{-ax}/x) \cdot x \cdot \cos(ux) dx = \int_0^{\infty} e^{-ax} \cos(ux) dx = F_c(u)$
S3	We know $F_c(u) = a/(a^2+u^2)$. So: $d/du [F_s(u)] = a/(a^2+u^2) \text{ Integrating: } F_s(u) = \int a/(a^2+u^2) du = \tan^{-1}(u/a) + C$
S4	Find C: As $u \rightarrow 0$, $F_s(0) = \int_0^{\infty} (e^{-ax}/x) \cdot \sin(0) dx = 0$ So: $0 = \tan^{-1}(0) + C \Rightarrow C = 0$
RESULT	$F_s[e^{-ax}/x] = \tan^{-1}(u/a)$

NOTE: This technique of differentiating under the integral sign (Leibniz rule) is powerful for finding transforms of functions divided by x . Memorize this pattern!

5. PARSEVAL'S THEOREM & INTEGRAL EQUATIONS

Parseval's theorem is the energy conservation theorem for Fourier transforms. It is used to evaluate difficult definite integrals by relating them to simpler known transforms.

5.1 Parseval's Theorem (Complex Fourier Transform)

PARSEVAL'S THEOREM

$$\int_{-\infty}^{+\infty} |f(x)|^2 dx = (1/2\pi) \int_{-\infty}^{+\infty} |F(u)|^2 du$$

This states that the total energy of a signal computed in the time domain equals (up to a factor) the total energy computed in the frequency domain.

5.2 Parseval's Theorems for Cosine and Sine Transforms

Cosine Version

$$\int_0^{+\infty} [f(x)]^2 dx = (2/\pi) \int_0^{+\infty} [Fc(u)]^2 du$$

Sine Version

$$\int_0^{+\infty} [f(x)]^2 dx = (2/\pi) \int_0^{+\infty} [Fs(u)]^2 du$$

5.3 Application: Evaluating Integrals using Parseval's Theorem

Example: Evaluate $\int_0^\infty \sin^2(t)/t^2 dt$ using Parseval's theorem

S1	Relate to known Fourier transform From Example (Q4 in PDF): The Fourier transform of $f(x) = 1- x $ for $ x \leq 1$, 0 otherwise gives $F(u) = 4\sin^2(u/2)/u^2$
S2	Apply Parseval's theorem $\int_{-\infty}^{+\infty} [f(x)]^2 dx = (1/2\pi) \int_{-\infty}^{+\infty} F(u) ^2 du$ $\int_{-1}^{+1} (1- x)^2 dx = (1/2\pi) \int_{-\infty}^{+\infty} [4\sin^2(u/2)/u^2]^2 du$
S3	Evaluate left side: $\int_{-1}^{+1} (1- x)^2 dx = 2 \int_0^1 (1-x)^2 dx = 2[-(1-x)^3/3]_0^1 = 2/3$
S4	So: $2/3 = (1/2\pi) \int_{-\infty}^{+\infty} 16\sin^4(u/2)/u^4 du = (8/\pi) \int_0^\infty \sin^4(u/2)/u^4 du$ Put $u/2 = t$: $u = 2t$, $du = 2dt$ $2/3 = (8/\pi) \int_0^\infty \sin^4(t)/(2t)^4 \cdot 2dt = (8/\pi) \int_0^\infty \sin^4(t)/8t^4 dt = (1/\pi) \int_0^\infty \sin^4(t)/t^4 dt$

Example: Solve the Integral Equation $\int_0^\infty f(x)\cos(\alpha x)dx = 1-\alpha$ ($0 \leq \alpha \leq 1$), 0 ($\alpha > 1$)

Integral equations where $f(x)$ is unknown are solved by recognizing the transform structure.

S1	Recognize the structure: the LHS is $Fc[\alpha] = 1-\alpha$ for $0 \leq \alpha \leq 1$, 0 otherwise
S2	Apply inverse cosine transform to find $f(x)$ $f(x) = (2/\pi) \int_0^\infty Fc(\alpha) \cdot \cos(\alpha x) d\alpha = (2/\pi) \int_0^1 (1-\alpha)\cos(\alpha x) d\alpha + 0$
S3	Integrate by parts with $p=(1-\alpha)$, $dq=\cos(\alpha x)d\alpha$

	$= (2/\pi)\{[(1-\alpha)\sin(\alpha x)/x]_0^1 + (1/x)\int_0^1 \sin(\alpha x)d\alpha\} = (2/\pi)\{0 + (1/x)[- \cos(\alpha x)/x]_0^1\} = (2/\pi \cdot x^2)(1 - \cos x) = (4/\pi x^2)\sin^2(x/2)$
SOLUTION	$f(x) = 4\sin^2(x/2) / (\pi x^2)$
S4	Deduce the integral: Put $\alpha=0$, $F_c[0] = 1$: $\pi \cdot 1 = 2 \int_0^\infty \sin^2(t)/t^2 dt \Rightarrow \int_0^\infty \sin^2(t)/t^2 dt = \pi/2$
CLASSIC INTEGRAL	$\int_0^\infty \sin^2(t)/t^2 dt = \pi/2$

6. CONVOLUTION THEOREM

Convolution is a mathematical operation that combines two functions to produce a third function representing how one function modifies the other. In Fourier analysis, convolution in the time domain corresponds to multiplication in the frequency domain — this is the Convolution Theorem.

6.1 Definition of Convolution

CONVOLUTION

$$(f * g)(x) = \int_{-\infty}^{+\infty} f(t) \cdot g(x-t) dt = \int_{-\infty}^{+\infty} f(x-t) \cdot g(t) dt$$

Convolution is commutative: $f * g = g * f$

6.2 Convolution Theorem for Fourier Transform

CONVOLUTION THEOREM

$$F[f * g] = F(u) \cdot G(u) \quad \text{OR} \quad F[f(x) \cdot g(x)] = (1/2\pi) (F * G)(u)$$

This theorem has tremendous practical importance:

- In signal processing: the output of an LTI system is the convolution of input and impulse response
- Filtering: multiplication in frequency domain achieves convolution filtering in time domain
- Simplifies calculation: convolution integrals become simple products after transformation

6.3 Worked Example using Convolution

Verify the Convolution Theorem for $f(x) = g(x) = e^{-x}$ ($x \geq 0$), 0 ($x < 0$)

S1	Compute convolution directly $(f * g)(x) = \int_{-\infty}^{+\infty} f(t)g(x-t)dt = \int_0^x e^{-t} \cdot e^{-(x-t)}dt$ (both t and $x-t$ must be ≥ 0) = $\int_0^x e^{-t} \cdot e^{-x+t}dt = e^{-x} \int_0^x dt = x \cdot e^{-x}$
S2	Fourier transform of convolution result: $F[xe^{-x}]$ Using known result or integration by parts: $F[xe^{-x}] = 1/(1+iu)^2$
S3	Compute $F(u)$ and $G(u)$ separately $F(u) = G(u) = F[e^{-x}u(x)] = \int_0^{+\infty} e^{-x}e^{iux}dx = \int_0^{+\infty} e^{x(-1+iu)}dx = 1/(1-iu)$
S4	Verify: $F(u) \cdot G(u) = 1/(1-iu)^2$ Check: $F[x \cdot e^{-x}] = 1/(1-iu)^2$ ✓ (consistent with step 2, noting sign convention difference)
VERIFIED	$F[f * g] = F(u) \cdot G(u)$ --- Convolution theorem confirmed

7. SELF-RECIPROCAL FUNCTIONS UNDER FOURIER TRANSFORM

A function $f(x)$ is called self-reciprocal (or self-dual) under the Fourier transform if its Fourier transform equals the function itself (up to a scalar factor). This means applying the Fourier transform to $f(x)$ gives back $f(x)$ (possibly rescaled).

SELF-RECIPROCAL CONDITION

$$F[f(x)] = F(u) = c \cdot f(u) \quad (\text{for some constant } c)$$

7.1 The Gaussian Function is Self-Reciprocal

The most important example of a self-reciprocal function is the Gaussian $e^{-a^2 x^2}$.

Proof: $F[e^{-a^2 x^2}] = (\sqrt{\pi/a}) e^{-u^2/(4a^2)}$

S1	Write the Fourier Transform $F(u) = \int_{-\infty}^{+\infty} e^{-a^2 x^2} \cdot e^{iux} dx = \int_{-\infty}^{+\infty} e^{-a^2 x^2 + iux} dx$
S2	Complete the square in the exponent $-a^2 x^2 + iux = -a^2 [x^2 - (iu/a^2)x] = -a^2 [(x - iu/(2a^2))^2 + (iu/(2a^2))^2 \text{ wait...}] = -a^2 (x - iu/(2a^2))^2 - u^2/4a^2$
S3	Factor out the constant $F(u) = e^{-u^2/(4a^2)} \int_{-\infty}^{+\infty} e^{-a^2 (x - iu/(2a^2))^2} dx$
S4	Substitute $t = a(x - iu/(2a^2))$, $dt = a \cdot dx$, limits still $-\infty$ to $+\infty$ $= e^{-u^2/(4a^2)} \cdot (1/a) \int_{-\infty}^{+\infty} e^{-t^2} dt = e^{-u^2/(4a^2)} \cdot (\sqrt{\pi/a})$
RESULT	$F[e^{-a^2 x^2}] = (\sqrt{\pi/a}) \cdot e^{-u^2/(4a^2)}$

Special Case: $a^2 = 1/2$ gives Self-Reciprocal

Put $a^2 = 1/2$ (so $a = 1/\sqrt{2}$):

$$F[e^{-x^2/2}] = (\sqrt{\pi} / (1/\sqrt{2})) \cdot e^{-u^2/2} = \sqrt{2\pi} \cdot e^{-u^2/2}$$

Since $f(x) = e^{-x^2/2}$ and $F(u) = \sqrt{2\pi} \cdot e^{-u^2/2} = \sqrt{2\pi} \cdot f(u)$, the Gaussian is self-reciprocal (up to factor $\sqrt{2\pi}$).

SELF-RECIPROCAL

$e^{-x^2/2}$ is self-reciprocal under complex Fourier transform

IMPORTANT: The fact that $e^{-x^2/2}$ is its own Fourier transform (up to constant) is fundamental in quantum mechanics (Hermite functions), probability (normal distribution), and signal processing (minimum uncertainty wavepackets). VTU asks this as a proof question.

8. DISCRETE FOURIER TRANSFORM (DFT)

The Discrete Fourier Transform extends Fourier analysis to finite sequences of discrete data points. It is the foundation of digital signal processing (DSP), image compression (JPEG), audio processing (MP3), and countless other applications. While the continuous Fourier transform processes analog signals, the DFT processes digital (sampled) signals.

8.1 Definition of DFT

Given a finite sequence of N complex numbers $x[0], x[1], x[2], \dots, x[N-1]$, the DFT produces another sequence $X[0], X[1], \dots, X[N-1]$:

DFT DEFINITION

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot W_N^{nk} \quad \text{where } W_N = e^{-j2\pi/N}$$

Here $W_N = e^{-j2\pi/N}$ is called the N -th root of unity or the twiddle factor. The complete DFT formula is:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j2\pi nk/N} \quad \text{for } k = 0, 1, 2, \dots, N-1$$

NOTE: In engineering notation: $j = \sqrt{-1}$ (to avoid confusion with current i). The summation runs over n from 0 to $N-1$, and we compute $X[k]$ for each k from 0 to $N-1$.

8.2 Inverse DFT (IDFT)

INVERSE DFT

$$x[n] = (1/N) \sum_{k=0}^{N-1} X[k] \cdot e^{j2\pi nk/N} \quad \text{for } n = 0, 1, \dots, N-1$$

8.3 The Twiddle Factor W_N

The twiddle factor $W_N = e^{-j2\pi/N}$ has very important properties that make DFT computation efficient:

Property	Formula
Periodicity	$W_N^{n+N} = W_N^n$ (periodic with period N)
Symmetry	$W_N^{n+N/2} = -W_N^n$
Special Values	$W_N^0 = 1, \quad W_N^N = 1$
Conjugate	$(W_N^{nk})^* = W_N^{-nk}$
Unity	$W_N^{nN} = 1$ for all integer n

8.4 Matrix Form of DFT

The DFT can be written as a matrix multiplication: $X = W \cdot x$, where W is the DFT matrix:

For $N = 4$:

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \begin{bmatrix} W^0 & W^0 & W^0 & W^0 \\ W^0 & W^1 & W^2 & W^3 \\ W^0 & W^2 & W^4 & W^6 \\ W^0 & W^3 & W^6 & W^9 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

where $W = W_4 = e^{-j2\pi/4} = e^{-j\pi/2} = -j$ (for $N=4$)

8.5 Worked Examples — DFT

Example 1: Compute the DFT of the sequence $x = \{1, 2, 3, 4\}$

This is the most common 4-point DFT example in VTU exams.

S1	Identify parameters: $N = 4$, so $W = W_4 = e^{-j\pi/2} = -j$ Powers of W : $W^0=1$, $W^1=-j$, $W^2=-1$, $W^3=j$, $W^4=1$ (cyclic)
S2	Compute $X[0]$ ($k=0$): sum of all elements $X[0] = x[0]W^0 + x[1]W^0 + x[2]W^0 + x[3]W^0 = (1+2+3+4) \cdot 1 = 10$
S3	Compute $X[1]$ ($k=1$): multiply $x[n]$ by W^n $X[1] = x[0]W^0 + x[1]W^1 + x[2]W^2 + x[3]W^3 = 1 \cdot 1 + 2 \cdot (-j) + 3 \cdot (-1) + 4 \cdot j = 1 - 2j - 3 + 4j = -2 + 2j$
S4	Compute $X[2]$ ($k=2$): multiply $x[n]$ by W^{2n} $X[2] = x[0]W^0 + x[1]W^2 + x[2]W^4 + x[3]W^6 = 1 \cdot 1 + 2 \cdot (-1) + 3 \cdot 1 + 4 \cdot (-1) = 1 - 2 + 3 - 4 = -2$
S5	Compute $X[3]$ ($k=3$): multiply $x[n]$ by W^{3n} $X[3] = x[0]W^0 + x[1]W^3 + x[2]W^6 + x[3]W^9 = 1 \cdot 1 + 2 \cdot j + 3 \cdot (-1) + 4 \cdot (-j) = 1 + 2j - 3 - 4j = -2 - 2j$

$X[0]$	$X[1]$	$X[2]$	$X[3]$
10	$-2+2j$	-2	$-2-2j$

NOTE: Notice: $X[0]$ is always the sum of all elements (DC component). $X[2] = -2$ is real, $X[1]$ and $X[3]$ are complex conjugates of each other. This conjugate symmetry holds for all real input sequences.

Example 2: Compute the 4-point DFT of $x = \{1, 0, 1, 0\}$

This sequence represents a sampled cosine wave — note how the DFT reveals the frequency content.

S1	$W = e^{-j\pi/2} = -j$. Powers: $W^0=1$, $W^1=-j$, $W^2=-1$, $W^3=j$
-----------	--

S2	$X[0] = 1+0+1+0 = 2$
S3	$X[1] = 1 \cdot 1 + 0 \cdot (-j) + 1 \cdot (-1) + 0 \cdot j = 1+0-1+0 = 0$
S4	$X[2] = 1 \cdot 1 + 0 \cdot (-1) + 1 \cdot 1 + 0 \cdot (-1) = 1+0+1+0 = 2$
S5	$X[3] = 1 \cdot 1 + 0 \cdot j + 1 \cdot (-1) + 0 \cdot (-j) = 1+0-1+0 = 0$
DFT RES ULT	$x = \{2, 0, 2, 0\}$

NOTE: Non-zero values at $k=0$ and $k=2$ tell us the sequence has frequency components at DC ($k=0$) and at the Nyquist frequency ($k=2$). This is exactly right for a sampled cosine at half the sampling frequency.

Example 3: Find the IDFT of $X = \{4, 1+j, 0, 1-j\}$

Inverse DFT: $x[n] = (1/N) \sum_{k=0}^{N-1} X[k] \cdot e^{j2\pi nk/N}$. For $N=4$, $W^{-1} = +j$.

Note: For IDFT, twiddle factor = $W^{-nk} = e^{+j2\pi nk/N}$. Powers: $(W^{-1})^0=1$, $(W^{-1})^1=j$, $(W^{-1})^2=-1$, $(W^{-1})^3=-j$

S1	$x[0] = (1/4)[X[0]+X[1]+X[2]+X[3]] = (1/4)[4+(1+j)+0+(1-j)] = (1/4)(6) = 3/2$
S2	$x[1] = (1/4)[X[0] \cdot 1 + X[1] \cdot j + X[2] \cdot (-1) + X[3] \cdot (-j)] = (1/4)[4 + j(1+j) + 0 - j(1-j)] = (1/4)[4 + j - 1 - j + 1] = (1/4)[4] = 1$
S3	$x[2] = (1/4)[4 + (1+j)(-1) + 0 + (1-j)(-1)] = (1/4)[4-1-j-1+j] = (1/4)(2) = 1/2$
S4	$x[3] = (1/4)[4 + (1+j)(-j) + 0 + (1-j)(j)] = (1/4)[4 + (-j+1) + (j+1)] = (1/4)(6) = 3/2$
IDFT RES ULT	$x = \{3/2, 1, 1/2, 3/2\}$

9. FAST FOURIER TRANSFORM (FFT)

The Fast Fourier Transform (FFT) is NOT a different transform from DFT — it is an efficient ALGORITHM to compute the DFT. The direct DFT computation requires $O(N^2)$ operations. The FFT reduces this to $O(N \log_2 N)$, making it vastly faster for large N .

Direct DFT Computation Multiplications needed: N^2 For $N=1024$: 1,048,576 operations Complexity: $O(N^2)$	FFT Algorithm Multiplications needed: $(N/2)\log_2(N)$ For $N=1024$: 5,120 operations Complexity: $O(N \log N)$
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9.1 Decimation in Time (DIT) FFT — Cooley-Tukey Algorithm

The most common FFT algorithm splits the N -point DFT into two $(N/2)$ -point DFTs of even-indexed and odd-indexed samples.

IDEA	Divide and Conquer Principle For $N = 2^m$ (power of 2), split $x[n]$ into: Even-indexed: $x[0], x[2], x[4], \dots, x[N-2]$ Odd-indexed: $x[1], x[3], x[5], \dots, x[N-1]$ Let $G[k]$ = DFT of even-indexed samples Let $H[k]$ = DFT of odd-indexed samples
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BUTTERFLY FORMULA	$X[k] = G[k] + W_N^k \cdot H[k] \quad \text{AND} \quad X[k+N/2] = G[k] - W_N^k \cdot H[k]$
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The computation $W_N^k \cdot H[k]$ is called a 'butterfly' operation. Each level of recursion has $N/2$ butterflies, and there are $\log_2(N)$ levels total, giving $(N/2)\log_2(N)$ total operations.

9.2 FFT Worked Example: 8-Point DFT using FFT Algorithm

Compute the 8-point DFT of $x = \{1, 1, 1, 1, 0, 0, 0, 0\}$ using FFT

This example demonstrates the FFT algorithm for $N=8$ ($= 2^3$, so 3 stages of butterflies).

S1	Identify $N=8$, $W_8 = e^{-j2\pi/8} = e^{-j\pi/4} = \cos(45^\circ) - j\sin(45^\circ) = (1-j)/\sqrt{2}$ Key twiddle factors: $W_8^0 = 1$, $W_8^1 = (1-j)/\sqrt{2} \approx 0.707 - 0.707j$, $W_8^2 = -j$, $W_8^3 = (-1-j)/\sqrt{2}$, $W_8^4 = -1$
S2	Stage 1: Split into even $\{x[0], x[2], x[4], x[6]\} = \{1, 1, 0, 0\}$ and odd $\{x[1], x[3], x[5], x[7]\} = \{1, 1, 0, 0\}$ 4-point DFT of even sequence $\{1, 1, 0, 0\}$: $G = \{2, 1-j, 0, 1+j\}$ 4-point DFT of odd sequence $\{1, 1, 0, 0\}$: $H = \{2, 1-j, 0, 1+j\}$
S3	Apply butterfly formula $X[k] = G[k] + W_8^k \cdot H[k]$ for $k=0, 1, 2, 3$

	$X[0] = G[0] + W^0 \cdot H[0] = 2 + 1 \cdot 2 = 4$ $X[1] = G[1] + W^1 \cdot H[1] = (1-j) + (0.707-0.707j)(1-j) = (1-j) + (0.707-0.707j-0.707j-0.707) = 1-j-1.414j = 1-2.414j$ $X[2] = G[2] + W^2 \cdot H[2] = 0 + (-j) \cdot 0 = 0$ $X[3] = G[3] + W^3 \cdot H[3] = (1+j) + ((-1-j)/\sqrt{2})(1+j)$
S4	Apply butterfly for $X[k+4] = G[k] - W_8^k \cdot H[k]$ $X[4] = G[0] - W^0 \cdot H[0] = 2 - 2 = 0$ $X[5] = G[1] - W^1 \cdot H[1] = (1-j) - (0.707-0.707j)(1-j) = 1+0.414j$ $X[6] = G[2] - W^2 \cdot H[2] = 0$ $X[7] = G[3] - W^3 \cdot H[3] = (1+j) - W^3(1+j)$
FFT RESULT	$x = \{4, 1-2.414j, 0, 1+0.414j, 0, 1-0.414j, 0, 1+2.414j\}$

9.3 Comparison: DFT vs FFT

N (Points)	DFT Operations (N^2)	FFT Operations ($(N/2)\log_2 N$)
8	64	12
64	4,096	192
256	65,536	1,024
1024	1,048,576	5,120
4096	16,777,216	24,576

IMPORTANT: The FFT algorithm is considered one of the most important algorithms of the 20th century. It enabled real-time signal processing, voice recognition, image compression (JPEG), mobile communications, MRI imaging, and many other technologies.

10. PREVIOUS YEAR VTU QUESTIONS WITH DETAILED SOLUTIONS

NOTE: These questions are from actual VTU examination papers. Master these and you are exam-ready!

10.1 May/June 2019 — Question 1 (10 Marks)

Find the Fourier transform of $f(x) = 1-x^2$ for $|x| < 1$, 0 for $|x| \geq 1$. Hence evaluate $\int_0^\infty (x \cos x - \sin x)/x^3 dx$ and $\int_0^\infty (x \cos x - \sin x)/x^3 \cos(x/2) dx$.

STEP 1	Write piecewise: $f(x) = 1-x^2$ for $-1 \leq x \leq 1$, 0 otherwise This is an EVEN function since $(1-(-x)^2) = 1-x^2 = f(x)$
STEP 2	Apply Fourier Transform (using even function property) $F(u) = \int_{-\infty}^{+\infty} f(x)e^{iux} dx = \int_{-1}^{+1} (1-x^2)e^{iux} dx$
STEP 3	Integrate by parts (three times for the x^2 term) $= [(1-x^2)e^{iux}/(iu)]_{-1}^{+1} + (1/iu) \int_{-1}^{+1} (-2x)e^{iux} dx = 0 + (-2/iu)[xe^{iux}/(iu)]_{-1}^{+1} + (2/iu) \int_{-1}^{+1} e^{iux} dx \cdot 1/(iu) = (-2/(iu)^2)[xe^{iux}]_{-1}^{+1} - (2/(iu)^3)[e^{iux}]_{-1}^{+1} = (2/u^2)[e^{iu} + e^{-iu}] - (2/iu^3)[e^{iu} - e^{-iu}] \dots = (4 \cos u)/u^2 - (4 \sin u)/u^3 = 4(ucos u - sin u)/u^3$
TRANSFORM	$F(u) = -4(ucos u - sin u)/u^3 = 4(sin u - ucos u)/u^3$
STEP 4	Evaluate $\int_0^\infty (x \cos x - \sin x)/x^3 dx$ using inverse transform at $x=0$ $f(0) = 1$. By inverse Fourier theorem: $1 = (1/2\pi) \int_{-\infty}^{+\infty} F(u) du = (1/2\pi) \int_{-\infty}^{+\infty} -4(ucos u - sin u)/u^3 du \cdot 2\pi = -4 \int_{-\infty}^{+\infty} (ucos u - sin u)/u^3 du = -8 \int_0^\infty (ucos u - sin u)/u^3 du \int_0^\infty (x \cos x - \sin x)/x^3 dx = -\pi/4$
RESULT (i)	$\int_0^\infty (x \cos x - \sin x)/x^3 dx = -\pi/4$
STEP 5	For second integral: use $x = 1/2$ in inverse transform $f(1/2) = 1 - (1/2)^2 = 3/4$. By inverse Fourier theorem: $3/4 = (1/2\pi) \int_{-\infty}^{+\infty} F(u)e^{-iu/2} du$ This gives $\int_0^\infty (ucos u - sin u)/u^3 \cdot \cos(u/2) du = -3\pi/16$
RESULT (ii)	$\int_0^\infty (x \cos x - \sin x)/x^3 \cos(x/2) dx = -3\pi/16$

10.2 Dec 2018/Jan 2019 — Question 2 (10 Marks)

Find the Fourier Cosine and Sine transform of $e^{-ax} \cdot \cos(ax)$.

S1	Cosine Transform: $F_c(u) = \int_0^\infty e^{-ax} \cos(ax) \cos(ux) dx$ Use product-to-sum: $\cos(A)\cos(B) = (1/2)[\cos(A-B) + \cos(A+B)]$ $\cos(ax)\cos(ux) = (1/2)[\cos((a-u)x) + \cos((a+u)x)]$
S2	Apply $F_c[e^{-ax}] = a/(a^2+u^2)$ to each term

	$F_c(u) = (1/2) \int_0^\infty e^{-ax} \cos((a-u)x) dx + (1/2) \int_0^\infty e^{-ax} \cos((a+u)x) dx = (1/2) \cdot a/(a^2+(a-u)^2) + (1/2) \cdot a/(a^2+(a+u)^2)$
Cosine Result	$F_c(u) = (a/2) [1/(a^2+(a-u)^2) + 1/(a^2+(a+u)^2)]$
S3	Sine Transform: use $\sin(A)\cos(B) = (1/2)[\sin(A+B)+\sin(A-B)]$ $F_s(u) = \int_0^\infty e^{-ax} \cos(ax) \sin(ux) dx \cos(ax) \sin(ux) = (1/2)[\sin((u+a)x) + \sin((u-a)x)] F_s(u) = (1/2) \cdot u+a / (a^2+(a+u)^2) + (1/2) \cdot u-a / (a^2+(a-u)^2)$
Sine Result	$F_s(u) = (1/2) [(u+a)/(a^2+(a+u)^2) + (u-a)/(a^2+(a-u)^2)]$

10.3 June/July 2017 — Question 3 (10 Marks)

Solve the integral equation: $\int_0^\infty F_c(\lambda) \cos(\lambda x) d\lambda = e^{-ax}$ ($a > 0$)

S1	Recognize: LHS is the inverse cosine transform of $F_c(\lambda)$ Comparing: $(2/\pi) \int_0^\infty F_c(\lambda) \cos(\lambda x) d\lambda = e^{-ax} \Rightarrow \int_0^\infty F_c(\lambda) \cos(\lambda x) d\lambda = (\pi/2) e^{-ax}$
S2	Take Fourier cosine transform of both sides $F_c[\int_0^\infty F_c(\lambda) \cos(\lambda x) d\lambda] = F_c[(\pi/2) e^{-ax}] (\pi/2) F_c(\lambda) = (\pi/2) \cdot a/(a^2+\lambda^2)$
SOLUTION	$F_c(\lambda) = a/(a^2 + \lambda^2)$

10.4 May/June 2018 — Question 4 (10 Marks)

Compute the 4-point DFT of the sequence $\{1, -1, 1, -1\}$. Verify using IDFT.

S1	$N=4, W_4=e^{-j\pi/2}=-j$. Powers: $W^0=1, W^1=-j, W^2=-1, W^3=j$
S2	$X[0] = 1+(-1)+1+(-1) = 0$
S3	$X[1] = 1+(-1)(-j)+1(-1)+(-1)(j) = 1+j-1-j = 0$
S4	$X[2] = 1+(-1)(-1)+1(1)+(-1)(-1) = 1+1+1+1 = 4$
S5	$X[3] = 1+(-1)(j)+1(-1)+(-1)(-j) = 1-j-1+j = 0$
DFT RESULT	$X = \{0, 0, 4, 0\}$

**VER
IFY****Apply IDFT: $x[n] = (1/4)\sum X[k]e^{j\pi nk/2}$** Only $X[2]=4$ is non-zero: $x[n] = (1/4) \cdot 4 \cdot e^{j\pi n \cdot 2/2} = e^{j\pi n} = \cos(\pi n) = (-1)^n \Rightarrow x = \{1, -1, 1, -1\}$ ✓ Verified!**10.5 Dec 2019/Jan 2020 — Question 5 (10 Marks)****Find the Fourier sine transform of $f(x)=e^{-ax}/x$. Hence show that $\int_0^\infty (e^{-ax}-e^{-bx})/x \, dx = \log(b/a)$.**

S1	We proved: $Fs[e^{-ax}/x] = \tan^{-1}(u/a)$ (See Section 4.3, Example 3)
S2	By linearity: $Fs[(e^{-ax}-e^{-bx})/x] = Fs[e^{-ax}/x] - Fs[e^{-bx}/x] = \tan^{-1}(u/a) - \tan^{-1}(u/b)$
S3	Apply inverse sine transform at $u \rightarrow 0^+$ (Parseval/limiting argument) Using the property that $Fs[f]$ at $u=0$ relates to $\int_0^\infty f(x)dx$: $\int_0^\infty (e^{-ax}-e^{-bx})/x \, dx = \lim_{u \rightarrow 0^+} \{ \pi/(2u) \} \cdot [\tan^{-1}(u/a) - \tan^{-1}(u/b)]$... applying L'Hopital or direct integration of $\tan^{-1}(u/a) - \tan^{-1}(u/b)$ from 0 to ∞ : $= (2/\pi) \int_0^\infty [\tan^{-1}(u/a) - \tan^{-1}(u/b)] \, du$
S4	Standard result: $\int_0^\infty \tan^{-1}(u/a) \, du = \pi a/2$ (integrating by parts) So: $(2/\pi)[\pi a/2 - \pi b/2]$? No — correct approach: Use the Fourier sine inversion at $x \rightarrow 0$: $\int_0^\infty f(x)dx = \lim_{x \rightarrow 0} \{ (\pi/2)f(x)/\dots \}$ instead apply: $\int_0^\infty (e^{-ax}-e^{-bx})/x \, dx = [\log(a^2+u^2)/2 \text{ from } 0 \text{ to } \infty]$...
S5	Direct approach using result $\int_0^\infty e^{-ax}dx = 1/a$: $\int_0^\infty (e^{-ax}-e^{-bx})/x \, dx = \int_0^\infty \int_a^b e^{-ux} \, du \, dx = \int_a^b [\int_0^\infty e^{-ux} \, dx] \, du = \int_a^b 1/u \, du = [\log u]_a^b = \log(b) - \log(a)$
FRUL LANI' S INTE GRA L	$\int_0^\infty (e^{-ax}-e^{-bx})/x \, dx = \log(b/a)$

11. QUICK REVISION SUMMARY

Use this section for last-minute revision before the VTU exam. All key results at a glance.

11.1 Transform Formulae — Complete Reference

Transform	Forward	Inverse
Fourier	$F(u) = \int_{-\infty}^{\infty} f(x) e^{iux} dx$	$f(x) = (1/2\pi) \int_{-\infty}^{\infty} F(u) e^{-iux} du$
Cosine	$F_c(u) = \int_0^{\infty} f(x) \cos(ux) dx$	$f(x) = (2/\pi) \int_0^{\infty} F_c(u) \cos(ux) du$
Sine	$F_s(u) = \int_0^{\infty} f(x) \sin(ux) dx$	$f(x) = (2/\pi) \int_0^{\infty} F_s(u) \sin(ux) du$
DFT	$X[k] = \sum_{n=0}^{N-1} x[n] W^{nk},$ $W = e^{-j2\pi/N}$	$x[n] = (1/N) \sum_{k=0}^{N-1} X[k] W^{-nk}$

11.2 Standard Transform Results — Memory Card

$F[f(x)]$	$F_c[f(x)]$	$F_s[f(x)]$
$F[e^{- x }] = 2/(1+u^2)$	$F_c[e^{-ax}] = a/(a^2+u^2)$	$F_s[e^{-ax}] = u/(a^2+u^2)$
$F[xe^{- x }] = 4iu/(1+u^2)^2$	$F_c[1-x, 0 \leq x \leq 1] = 2\sin^2(u/2)/u^2$	$F_s[e^{-ax}/x] = \tan^{-1}(u/a)$
$F[\text{rect}] = 2\sin(au)/u$	$F_c[1-x^2, -1 \leq x \leq 1] = 4(\sin u - u \cos u)/u^3$	$F_s[e^{- x }] = u/(1+u^2)$
$F[1- x] = 4\sin^2(u/2)/u^2$	$F_c[x, 0 \leq x \leq 1] = \cos^2(u/2)/u^2$ type	$F_s[x \cdot e^{-ax}] = 2au/(a^2+u^2)^2$
$F[e^{-a^2x^2}] = (\sqrt{\pi}/a) e^{-u^2/(4a^2)}$	$F_c[e^{-ax} \cos(ax)]:$ product formula	Parseval: $\int f ^2 = (2/\pi) \int F_s ^2$

11.3 Important Integrals to Remember

Integral	Value / Source
$\int_0^{\infty} \sin(x)/x dx$	$\pi/2 \leftarrow$ From rect function FT
$\int_0^{\infty} \sin^2(t)/t^2 dt$	$\pi/2 \leftarrow$ From triangular pulse FT
$\int_0^{\infty} x \cdot \sin(mx)/(1+x^2) dx$	$(\pi/2)e^{-m} \leftarrow$ From $F_s[e^{-x}]$
$\int_0^{\infty} \cos(mx)/(a^2+x^2) dx$	$(\pi/2a)e^{-am} \leftarrow$ From $F_c[e^{-ax}]$
$\int_0^{\infty} (x \cos x - \sin x)/x^3 dx$	$-\pi/4 \leftarrow$ From $F_t[1-x^2]$
$\int_{-\infty}^{+\infty} e^{-t^2} dt$	$\sqrt{\pi} \leftarrow$ Gaussian integral (fundamental)
$\int_0^{\infty} (e^{-ax} - e^{-bx})/x dx$	$\log(b/a) \leftarrow$ Frullani's integral

11.4 Exam Strategy — VTU Tips

- Always write the definition of the transform before starting computation
- Check if the function is even or odd FIRST — this determines which transform simplifies the problem
- For deducing integral values: use the inverse transform at $x=0$ or $x=m$ as appropriate
- For DFT: always compute twiddle factor W_N and all its needed powers before starting
- In FFT: state that it is Cooley-Tukey DIT algorithm. Draw the signal flow diagram if time permits
- Parseval's theorem: always equate energy on both sides carefully; handle factor of $2/\pi$ correctly
- Self-reciprocal: for the Gaussian, show the completing-the-square step explicitly
- For integral equations: identify it as inverse cosine or sine transform, then apply the transform to solve for $f(x)$

Best of luck for your VTU Examination! — vtu-easy.com

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